Data-Driven Prediction of AI-Training Structural Transitions

Target-conditioned location, scale, resolution landscapes and cross-temporal change

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Abstract

This paper develops a forecasting procedure for future changes in a measured structural description of AI training. It defines a cross-temporal displacement target, separates location and scale features, and constructs causal WPM landscapes and compatible probability forecasts. Proof trails show the effect of coordinate normalization, the limits of endpoint labels and the optimality criterion of a probability score. Worked examples distinguish a structural transition from failure or a change in model quality. Optional n/s descriptors are identified separately from compact observable proxies.

Research lineage and mathematical status. This study develops a WPM mathematical extension with reference to the author's published WPM architecture [1] and foundational n/s scaling-spectrum framework [2]. The n/s paper concerns conditional moment-spectrum parameterization; the WPM paper provides the broader architectural reference. The results proved here are mathematical consequences of stated assumptions, usually applications of standard linear algebra, probability or optimization. The WPM contribution is their organization into an explicit workflow. Modeling constructions and applicability conditions are identified separately; no universal forecasting superiority is asserted.

Predictor inputs. The forecasting layer described here is data driven: it learns from observed histories and matured outcomes. It is not fed the system's true governing equation or unobserved future states at issuance. Physical equations used below define observables, explain a limiting example or specify a separately labeled comparator. Optional supplied-map inverse problems and physics-informed variants must be declared separately; they are not evidence that the default forecast knows the generator.

1 Causal observation and resolution contract

Let $\mathcal{F}_t$ contain the observations actually reported by issuance t: loss measurements, gradient and update statistics, known schedules and verified context. A delayed validation measurement belongs to the information set only when reported. The forecast horizon P is fixed by the query; W selects past extent, and m is a declared temporal sampling or internal-resolution parameter. These variables need not be functions of one another. For each target T, issue a forecast $p_{s,P,r}$

using only $\mathcal{F}_s$, with lens $r = (W, m)$. Once the label has matured, its loss can enter a causal surface

$$\widehat{L}_t(r; T, P) = \frac{\sum_{s \in \mathcal{A}_t} w_s \ell_T(Y^T_{s,P}, p_{s,P,r})}{\sum_{s \in \mathcal{A}_t} w_s}. \tag{1.1}$$

The same matured set $\mathcal{A}_t$ is used in numerator and denominator. Extend maturity when persistence or reporting delay requires follow-up beyond $s + P$. Unresolved negative outcomes are not known negatives. Empty history is missing risk, not zero risk. A pocket is a connected low-risk region under a specified adjacency and threshold. Retain each pocket's flow cross-sections, position, existence, signed errors and identity history. Fixed uses an eligible reference lens; Multi aggregates same-target alternatives; Auto changes a specified policy with causal context and matured utility. A pocket minimum is not an automatic event probability. An interpretable candidate event model is

$$p_{t,P} = \operatorname{logit}^{-1}\left(\beta_0 + \beta^\top z_t + \sum_i \alpha_i \phi(\mathcal{P}_i) + \sum_{i<j} \gamma_{ij} \psi(\mathcal{P}_i, \mathcal{P}_j)\right). \tag{1.2}$$

The coefficients, feature maps and any calibration map must be fitted from permitted data. Pocket labels must not have an artificial ordinal meaning; use a permutation-invariant aggregation or preserve a well-defined relational graph. Shared observations create dependence among pockets and cannot be counted as extra independent measurements. The compact archived AI representation used observable location/scale proxies in its multi-system scenario comparison, not explicit fits of all n/s parameters. A full scaling-spectrum branch is an optional extension that must calculate and qualify its own descriptors. Structural location/scale features and an actually fitted n/s spectrum are not interchangeable descriptions.

2 Define the structural state before predicting its change

A structural state is a measured or estimated representation of training, not the entire parameter vector and not a semantic statement about capability. One compact state is $z_t = (\ell_t, b_t, c_t, d_t)$, containing clean-probe-loss level, local slope, curvature and a declared shift statistic. A scale state can contain increment dispersion, growth of dispersion and tail summaries. Every estimate uses the issuance prefix and its own units.

For example, fit $L_{t+u} \simeq a + bu + cu^2/2$ on $u \in [-W, 0]$ to define (a, b, c), and let d compare two causal subwindow levels. This is an observable descriptor model, not the true optimizer equation. Changing W changes the descriptor and can alter the meaning of a transition unless the target state convention is fixed separately from candidate forecast lenses.

Definition 2.1 (Cross-temporal displacement target)**.** Choose a fixed structural measurement map z, positive scaling matrix D, horizon P and threshold b. Define

$$\Delta_{t,P} = \left\| D^{-1}(z_{t+P} - z_t) \right\|, \qquad Y^{\mathrm{tr}}_{t,P} = \mathbf{1}\{\Delta_{t,P} > b\}. \tag{2.1}$$

The target map, scaling and threshold are specified physically or learned from permitted development data and then frozen for scoring.

This endpoint displacement does not automatically mean a first transition anywhere within the interval. For an arrival target, replace it explicitly by a path functional such as $\max_{1 \le h \le P} \|D^{-1}(z_{t+h} - z_t)\|$. Both are valid questions, but their labels and probability calibration differ.

3 Coordinate semantics and a normalization theorem

Level, slope and curvature have different units. Euclidean distance between them without a scale convention depends on the arbitrary unit of loss and time. A positive definite metric M gives $\Delta^2 = (z_{t+P} - z_t)^\top M(z_{t+P} - z_t)$.

Proposition 3.1 (Coordinate-invariant displacement under a matched metric)**.** *For an invertible linear coordinate map $u = Az$, choosing $M_u = A^{-\top} M A^{-1}$ preserves displacement exactly.*

Proof. Substitute $u_{t+P} - u_t = A(z_{t+P} - z_t)$ into the quadratic form and cancel $A^\top A^{-\top}$ and $A^{-1}A$. □

Using separate standard deviations is one diagonal metric choice. A full covariance metric accounts for correlated descriptors but requires stable estimation. Singular directions must be handled explicitly; regularization changes the metric. The metric is a definition of what constitutes structural change, not a theorem that normalized systems share future behavior.

Worked example 3.2 (A worked displacement)**.** Let $z_t = (0.4, -0.01)$, $z_{t+P} = (0.5, 0.01)$ and $D = \mathrm{diag}(0.1, 0.02)$. The standardized displacement is $(1, 1)$ with norm $\sqrt{2}$. At boundary b=1.2 the event occurs. Rescaling loss by 100 and its level/slope scales by 100 leaves this result unchanged. Rescaling only the observations changes the event incorrectly.

4 Why scale can contain information beyond location

Two recent trajectories can have similar estimated centers and different increment distributions. A larger conditional spread can raise the probability of leaving a structural neighborhood even when expected displacement remains small. This is a distributional statement, not an assertion that volatility always predicts failure.

Suppose a one-dimensional future displacement satisfies $Z \mid \mathcal{F}_t \sim N(\mu, s^2)$ and the event is $|Z| > b$. Then

$$p = \Phi\left(\frac{-b - \mu}{s}\right) + 1 - \Phi\left(\frac{b - \mu}{s}\right). \tag{4.1}$$

At $\mu = 0$, b=1, increasing s from 1/2 to 1 changes p from $2[1 - \Phi(2)]$ to $2[1 - \Phi(1)]$. Location alone cannot determine these probabilities. Under non-Gaussian errors, shape and tails matter too; a Gaussian location–scale map is an assumption to be checked.

5 Descriptor uncertainty and measurement error

The estimated $\widehat{z}_t$ contains measurement and fitting error. If $\widehat{z}_t = z_t + e_t$, then measured displacement is $z_{t+P} - z_t + e_{t+P} - e_t$. Its covariance includes $\mathrm{Cov}(e_{t+P}, e_t)$ when windows overlap. A threshold based on noisy displacement can label measurement noise as a transition.

A generative measurement model can propagate that uncertainty, or the operational target can explicitly be a transition of the measured descriptor. These are different tasks. Do not claim recovery of a latent training regime when only a noisy telemetry threshold is defined. Signed matured forecast errors, local residual bias and pocket disappearance may be additional causal features, but they are not independent measurements of a regime.

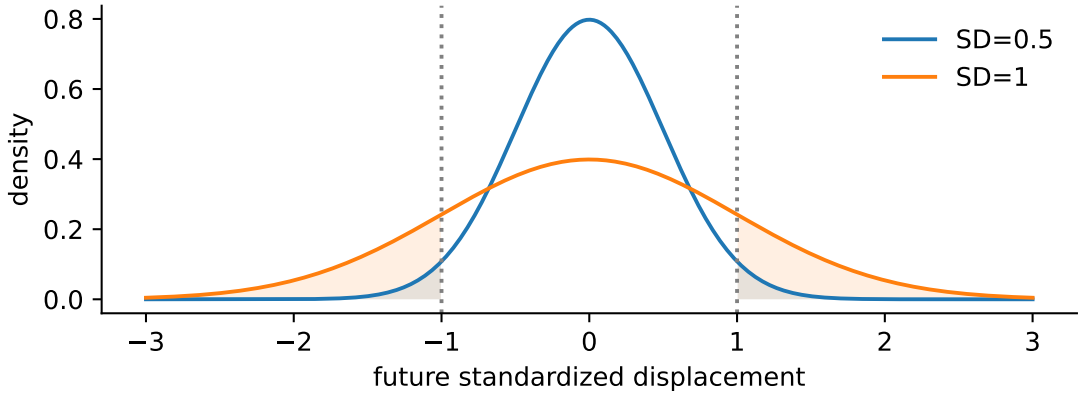


Figure 1: Equal-mean Gaussian displacement models with different scales assign different probabilities to leaving the interval [-1,1]. Shaded regions are analytical tails, not measured forecast performance.

6 From the landscape to a transition predictor

For each candidate lens, fit a model of $p_{t,P} = \mathbb{P}(Y^{\mathrm{tr}}_{t,P} = 1 \mid \mathcal{F}_t)$. The modeling class might be logistic regression, a conditional distribution for displacement or a separately qualified dynamical predictor. Build its landscape using the fixed transition target, not its current in-sample fitting residual.

Retain short-window and long-window pockets when they represent useful alternatives. A change in the lowest-risk cell can occur because the selector prefers another persistent basin; it is not a measured structural transition of training. The label Y^{tr} is defined by the structural state, independent of whichever lens wins.

If normalized weights π_i combine same-target probabilities, $p = \sum_i \pi_i p_i$ remains a probability. If a policy uses $\pi_i \propto e^{-\eta \widehat{L}_i}$, those weights are utility weights, not probabilities that a pocket is the true physical explanation. Reusing the same underlying history in many cells does not create independent votes.

7 Probability accuracy and ranking

Theorem 7.1 (The event probability minimizes Brier loss)**.** *For binary Y and $p_* = \mathbb{P}(Y = 1 \mid \mathcal{F}_t)$,*

$$\mathbb{E}[(Y - p)^2 \mid \mathcal{F}_t] = p_*(1 - p_*) + (p - p_*)^2. \tag{7.1}$$

Proof. Expand $p_*(1-p)^2 + (1-p_*)p^2$ and complete the square. The unique minimizer is p_*. □

A monotone probability calibrator can alter this loss while preserving ranking. Thus lower Brier and higher discrimination answer different questions. A learned transition score can be informative but miscalibrated after architecture, optimizer or event-prevalence changes [3].

8 Endpoint changes, transient excursions and failure

Consider a state starting at zero and moving to 2 before returning to zero at horizon 2. Its endpoint displacement is zero, but its maximum displacement is 2. A first-arrival label catches

the excursion; the endpoint label does not. Select the target according to the actual question rather than treating these as contradictory forecasts.

A transition is also not inherently harmful. A scheduled learning-rate reduction or convergence to a new stable regime may create structural displacement while improving the model. Failure requires a separate boundary or quality criterion. Changes in a scaling descriptor do not by themselves establish changes in semantic model competence or final generalization.

9 Optional n/s features and prospective response

If generalized moment spectra are actually computed and admissible, the released n/s family provides $H(q) = |q|^{n-1}/[s_0 e^{kq}]$. A descriptor $\Theta = (n, \log s_0, k)$ may then be used as an observed structural representation, with its fitting error and moment-admissibility flags. The logarithmic design $(\log|q|, 1, q)$ must have adequate rank and conditioning.

Forecasting future response from Θ_t still requires a learned temporal predictor. The spectrum equation relates moment orders at a given state; it is not automatically a time-evolution equation for $n(t), s_0(t), k(t)$. A response-magnitude target $\|\Theta_{\text{future}} - \Theta_{\text{current}}\|$ is also different from the compact proxy displacement defined earlier. Keep their evidence and labels separate.

10 An application worksheet

Suppose the required target is endpoint structural displacement over P=25 steps. Define the structural state map, loss units and metric using the observation protocol. Fit candidate predictors at independent W,m settings using prior complete examples. Store their actual issued probabilities. At time t, use only labels whose descriptor at $s + 25$ has been reported.

Extract pockets, preserving full flow and existence histories. Use a Fixed reference or a qualified same-target mixture. If a current structural descriptor is unavailable because a probe evaluation was delayed, do not fill it with the later value; either use a declared missing-feature model or abstain. Apply a separately fitted calibrator only in its supported context.

Return the target definition, horizon, probability, uncertainty interpretation and applicability state. A later refresh uses a new information set and is a new issuance. If the user instead asks whether any excursion will occur before a fixed terminal time, change the label and remaining horizon explicitly.

11 Telemetry comparison and applicability

Scoped evidence statement. In the archived compact multi-system comparison, structural location-plus-scale had lower Brier and stronger discrimination than the specified recent-telemetry comparator for this transition target [5]. These results use measured proxy features and a learned classifier; they are not evidence that a closed-form n/s equation was fed the optimizer's future law. A general statement that performance is merely "close to telemetry" would hide that target-specific difference. Similarity to telemetry was observed for other metrics or targets, not uniformly here.

The appropriate deployment rule remains conditional: compare matched-information structural, telemetry and combined predictors for the intended architecture, optimizer, horizon and target. Independent runs are the natural units when within-run windows overlap. A transition probability is not a recommended optimizer intervention unless a separate action-response model is supplied.

12 Rules and solved questions

1. Fix a structural measurement map and target scale independently of candidate lens selection.
2. Distinguish endpoint displacement, interval excursion, regime membership and failure.
3. Use only matured outcomes; keep location, scale and optional n/s fitting uncertainty explicit.
4. Preserve individual pockets without turning selection changes into physical state changes.
5. Return a probability and an applicability statement, not an unqualified claim of general model-quality prediction.

Exercise 12.1. A state starts at 0, reaches 2, and returns to 0. At boundary 1, compare endpoint and within-horizon transition labels.

Solution. The endpoint label is zero and the maximum-excursion label is one. Their predictors estimate different events.

Exercise 12.2. Does equal future mean displacement imply equal transition probability?

Solution. No. The Gaussian scale example gives different tail probabilities at the same mean. Non-Gaussian alternatives can differ even with the same variance.

Scope of the mathematical exposition

Unless explicitly identified as a sourced evidence statement, numerical values are constructed arithmetic examples. Application requires specifying the observation process, fitting unknown quantities, and checking reliability for the intended population and horizon. This study shares the AI prediction collection DOI 10.5281/zenodo.23784369.

References

[1] T. Nguyen. *The WPM Framework: A Generalized Mathematical Architecture for Cross-Scale, Spatial, and Dynamical Prediction*. Zenodo, 2026. doi:10.5281/zenodo.23248983.

[2] T. Nguyen. *An n/s Scaling-Spectrum Framework for Generalized Moment Scaling: Curvature, Signed-Moment Asymmetry, and Prospective Moment-Order Validation*. Zenodo, 2026. doi:10.5281/zenodo.23111783.

[3] T. Gneiting and A. E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association* 102(477), 359–378, 2007. doi:10.1198/016214506000001437.

[4] S. Boyd and L. Vandenberghe. *Introduction to Applied Linear Algebra: Vectors, Matrices, and Least Squares*. Cambridge University Press, 2018. https://stanford.edu/~boyd/vmls/.

[5] T. Nguyen. *Target-Conditioned Structural Forecasting of AI-Training Transitions: Frozen Multi-System Evidence for 1A–1D*. Version 1.1, October 2026. Archived manuscript, pp. 1–7. Evidence cited only for the scoped comparison statement.